YouTube Data Insights

YouTube Content Strategy Analysis

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Analysis Objectives

- Identify key drivers of views and engagement
- Determine optimal timing and content strategy
- Analyze high-ROI categories for advertisers
- Provide data-driven recommendations

Dataset Overview

YouTube Trending Videos Dataset

- Includes views, likes, comments, and engagement rate
- Covers multiple categories and creators
- Contains publish time (hour/day) and country-wise data
- Enables analysis of content trends and audience engagement

Tools & Technologies

- Python → Core language for data analysis
- Pandas → Data cleaning and manipulation
- Matplotlib → Basic data visualization
- Seaborn → Advanced and statistical visualizations
- Google Colab → Interactive analysis and presentation

Data Cleaning Steps

- Handled missing and inconsistent values
- Converted duration to numeric format (seconds) PT5M30S = 5 minutes 30 seconds
- Removed duplicates and irrelevant columns

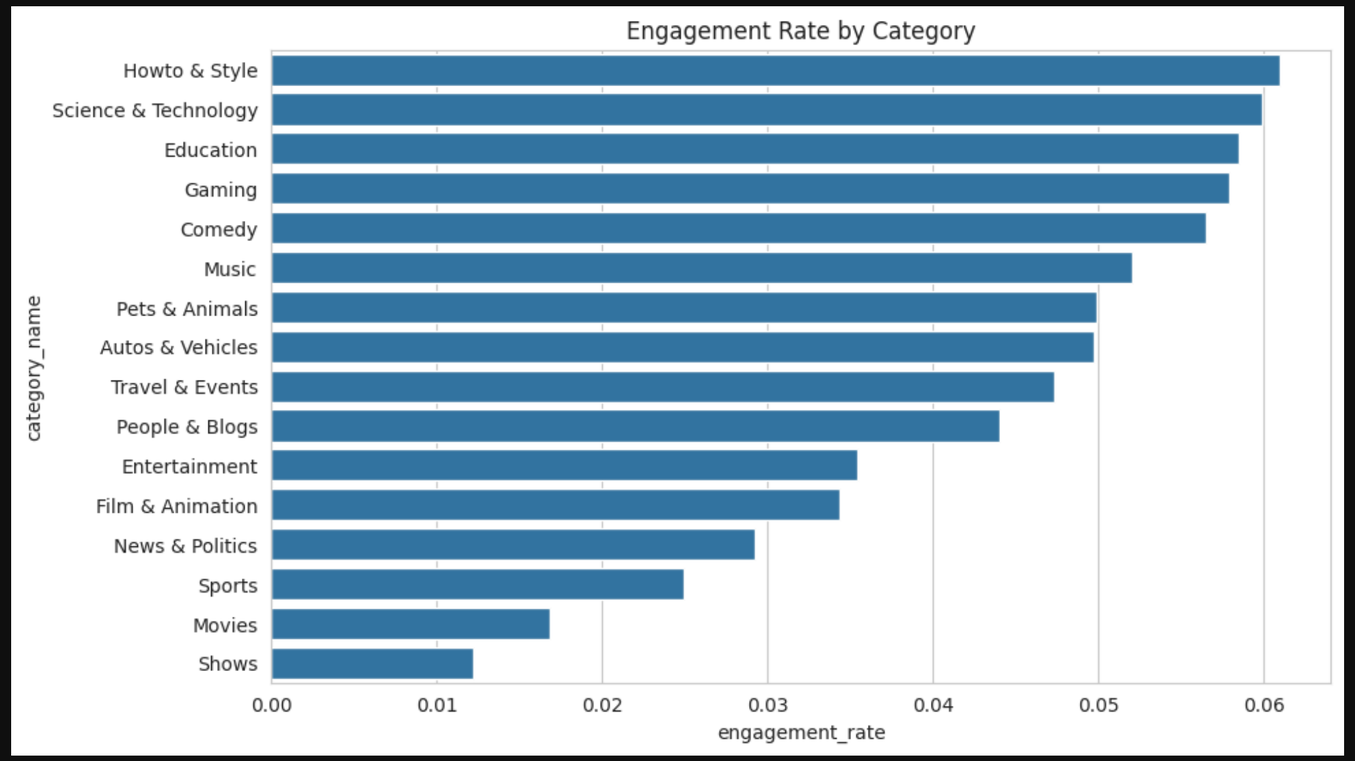
```
df['trending_date'] = pd.to_datetime(df['trending_date'], format='%y.%d.%m')
```

```
# remove nulls
df.dropna(inplace=True)
```

PART 1: CONTENT STRATEGY FOR CREATORS

Engagement Rate by Category

- How-to, Tech, and Education have the highest engagement rates
- Entertainment has high reach but relatively lower engagement quality
- Niche categories drive more loyal and interactive audiences

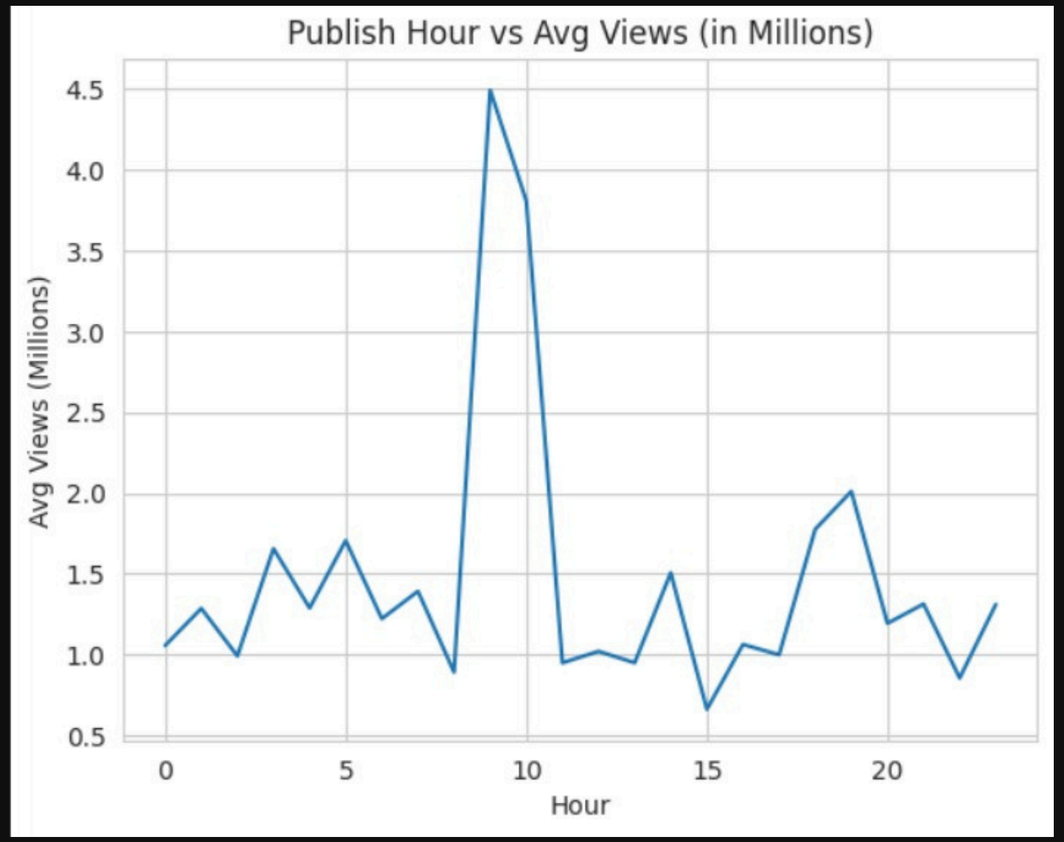



```
# create engagement metric
df['engagement'] = df['likes'] + df['comment_count']
```

```
category_engagement = (
    df.groupby('category_name')['engagement_rate']
    .mean()
    .sort_values(ascending=False)
    .reset_index()
)

plt.figure(figsize=(10,6))
sns.barplot(x='engagement_rate', y='category_name', data=category_engagement)
plt.title("Engagement Rate by Category")
plt.show()
```

Publish Hour vs Avg Views (in Millions)

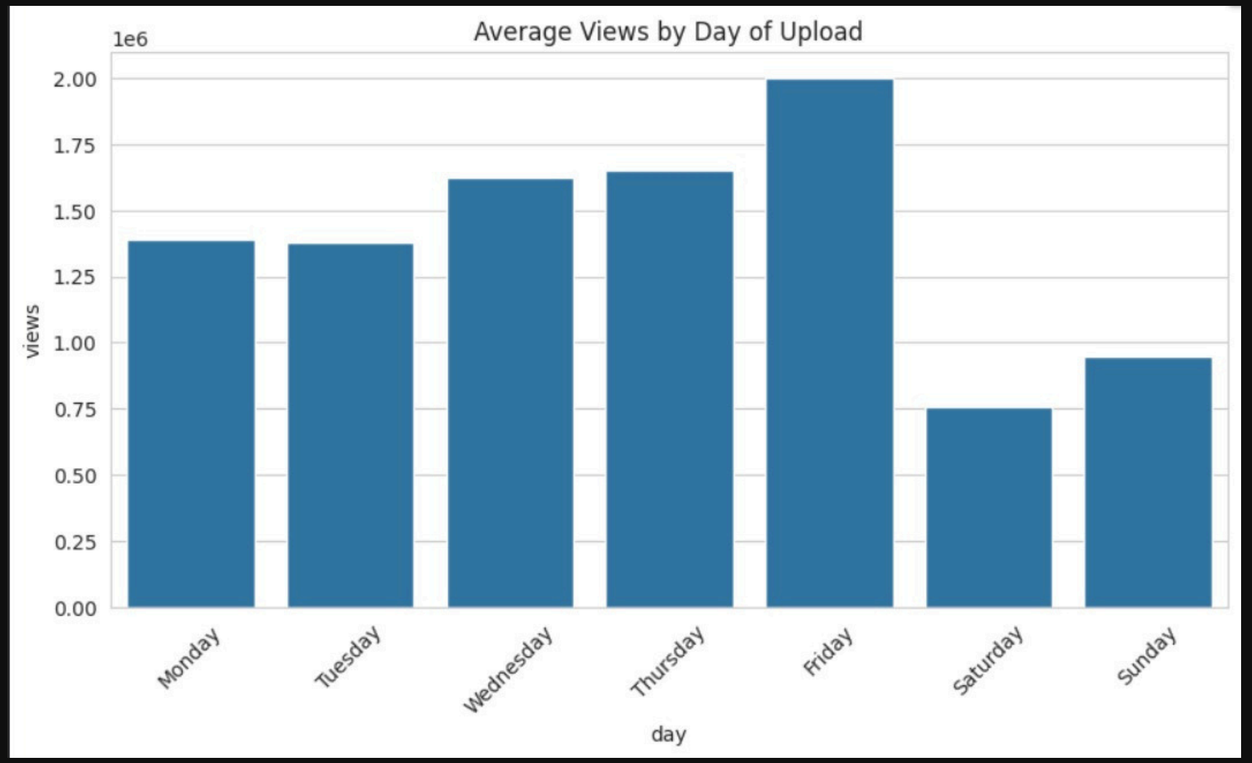
- Peak views occur around 9–10 AM (strong spike)
- Secondary engagement seen in evening hours (6–8 PM)
- Midday and late night show lower performance




```
df['publish_time'] = pd.to_datetime(df['publish_time'], utc=True)

df['publish_time_ist'] = df['publish_time'].dt.tz_convert('Asia/Kolkata')

df['publish_hour'] = df['publish_time_ist'].dt.hour
hour_views = df.groupby('publish_hour')['views'].mean() / 1e6

plt.figure()
sns.lineplot(x=hour_views.index, y=hour_views.values)
plt.title("Publish Hour vs Avg Views (in Millions)")
plt.xlabel("Hour")
plt.ylabel("Avg Views (Millions)")
plt.show()
```

Avg Views by Day of Upload

- Friday generates the highest average views
- Weekdays outperform weekends overall
- Saturday shows the lowest performance

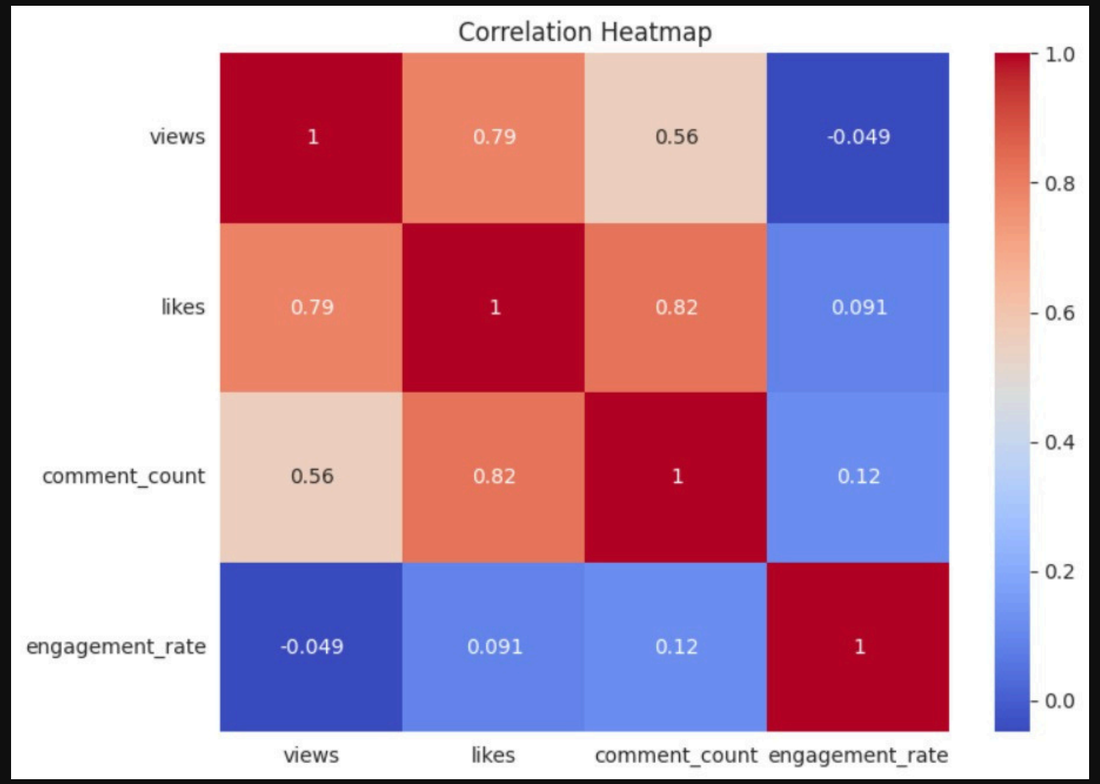



```
df['publish_time'] = pd.to_datetime(df['publish_time'], utc=True)
df['publish_time'] = df['publish_time'].dt.tz_convert('Asia/Kolkata')

df['day'] = df['publish_time'].dt.day_name()
```

```
day_views = df.groupby('day')['views'].mean().reindex([
    'Monday','Tuesday','Wednesday','Thursday','Friday','Saturday','Sunday'
]).reset_index()

plt.figure(figsize=(10,5))
sns.barplot(x='day', y='views', data=day_views)
plt.title("Average Views by Day of Upload")
plt.xticks(rotation=45)
plt.show()
```

Correlation Heatmap

- Strong correlation between likes and comments (0.82)
- Views are highly correlated with likes (0.79)
- Engagement rate has weak direct correlation with views

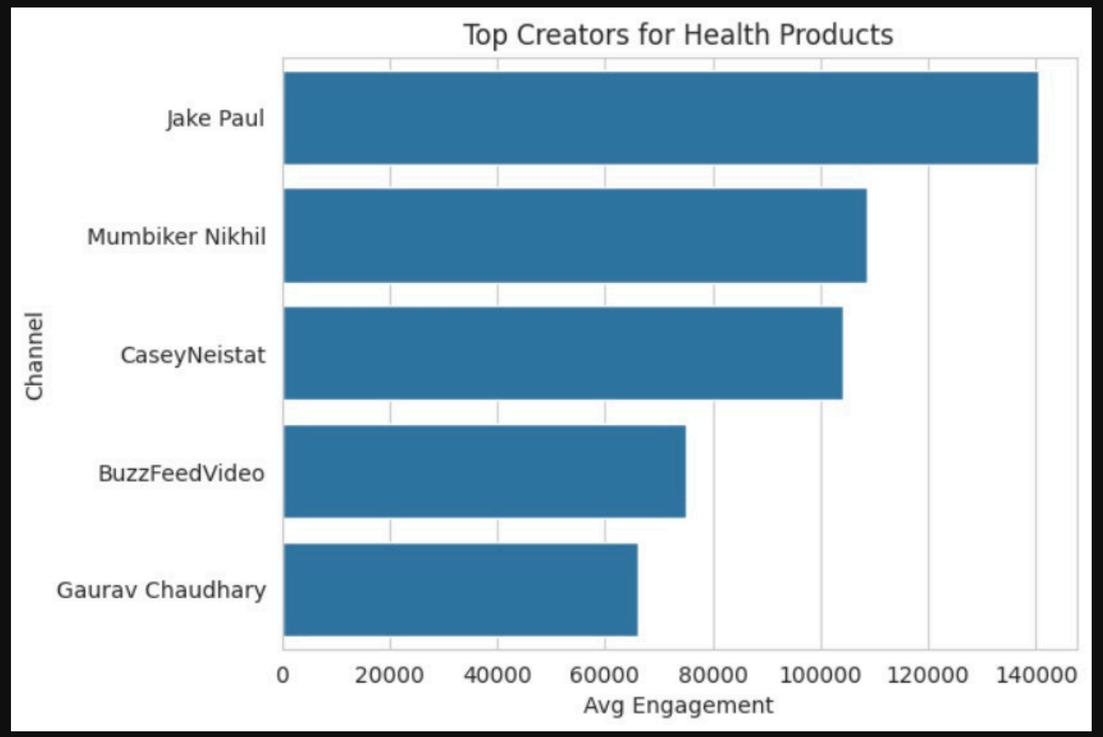


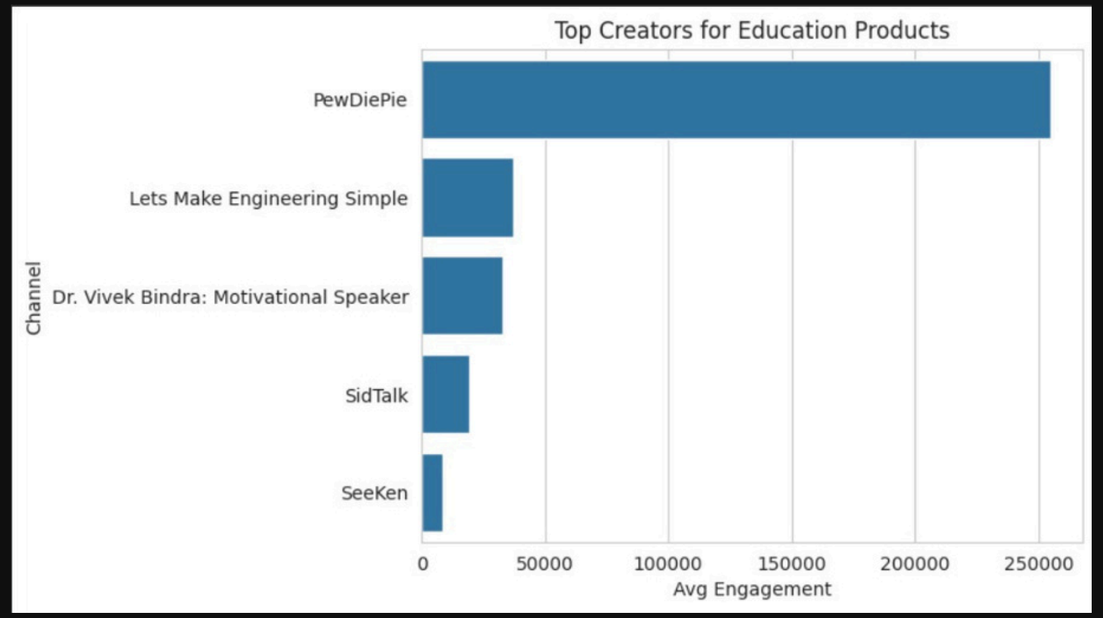

```
plt.figure(figsize=(8,6))

sns.heatmap(
    df[['views','likes','comment_count','engagement_rate']].corr(),
    annot=True,
    cmap='coolwarm'
)

plt.title("Correlation Heatmap")
plt.show()
```

PART 2: INVESTMENT STRATEGY FOR ADVERTISERS

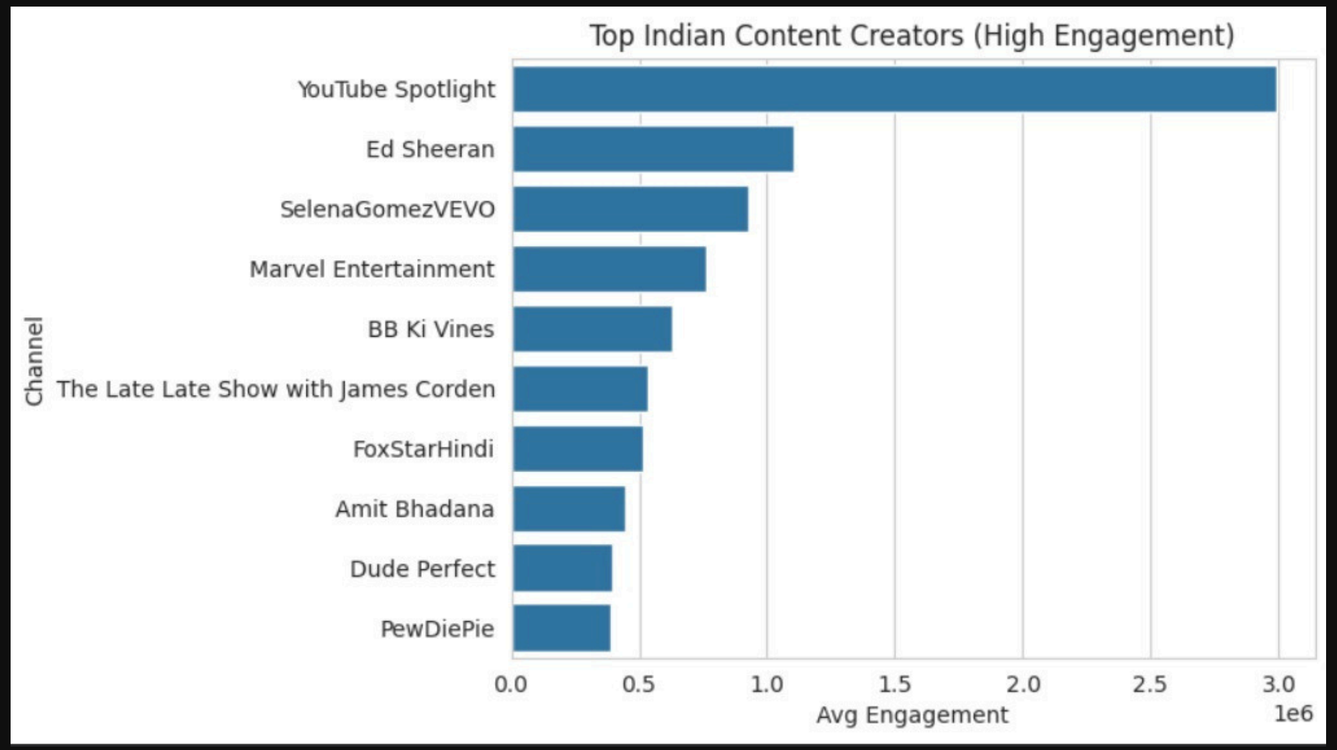
Top Creators for Product Categories

- Top creators dominate engagement → influencer power is highly concentrated
- Education creators have very high engagement despite smaller scale
- Lifestyle & fashion creators attract consistent audience interaction





Top Indian Content Creators (High Engagement)



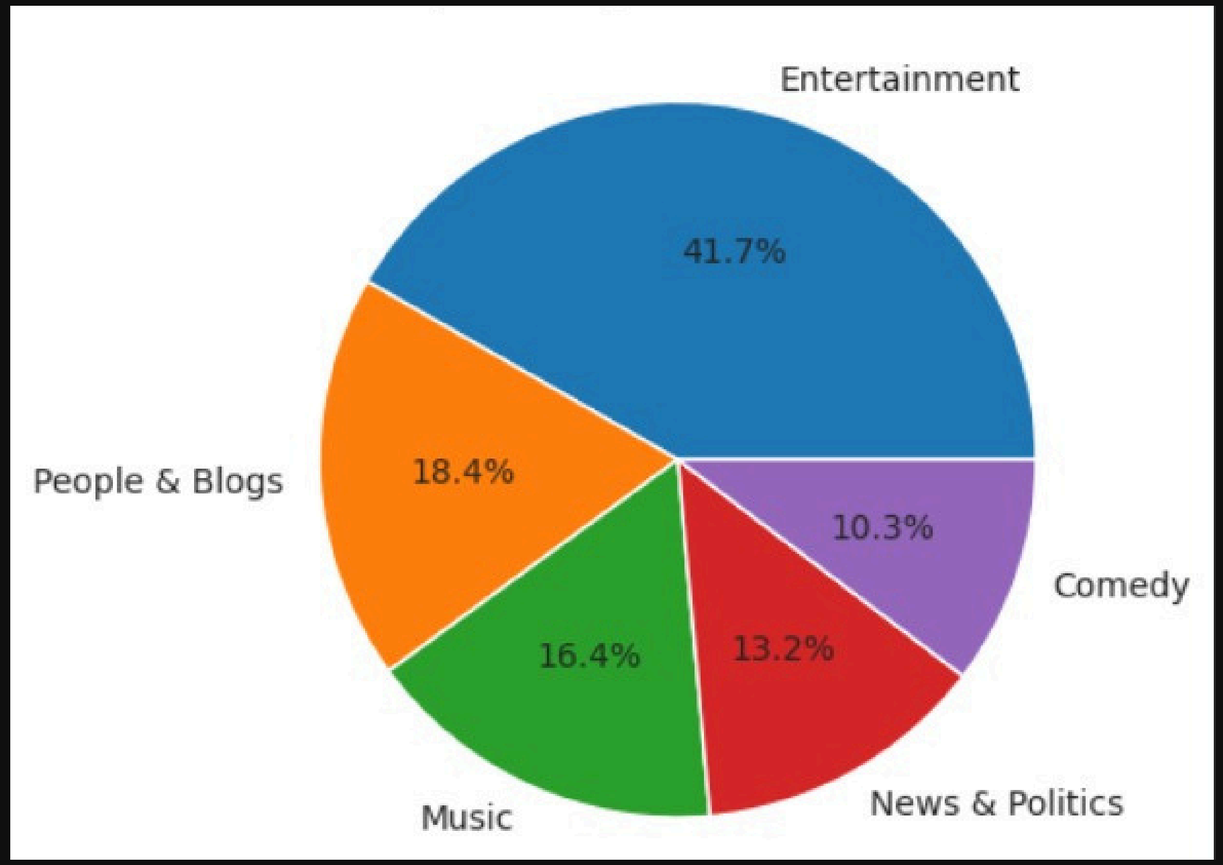
- Few creators (like YouTube Spotlight) massively outperform others
- Entertainment-based creators dominate high engagement rankings
- Engagement is skewed toward top-performing channels

```
india_df = df[df['country'] == 'IN']
```

```
india_df = df[df['country'] == 'IN'].copy()
india_df['engagement'] = india_df['likes'] + india_df['comment_count']
```

```
top_creators = (
    india_df.groupby('channel_title')['engagement']
    .mean()
    .sort_values(ascending=False)
    .head(10)
)

plt.figure()
sns.barplot(x=top_creators.values, y=top_creators.index)
plt.title("Top Indian Content Creators (High Engagement)")
plt.xlabel("Avg Engagement")
plt.ylabel("Channel")
plt.show()
```

Top 5 Categories Distribution



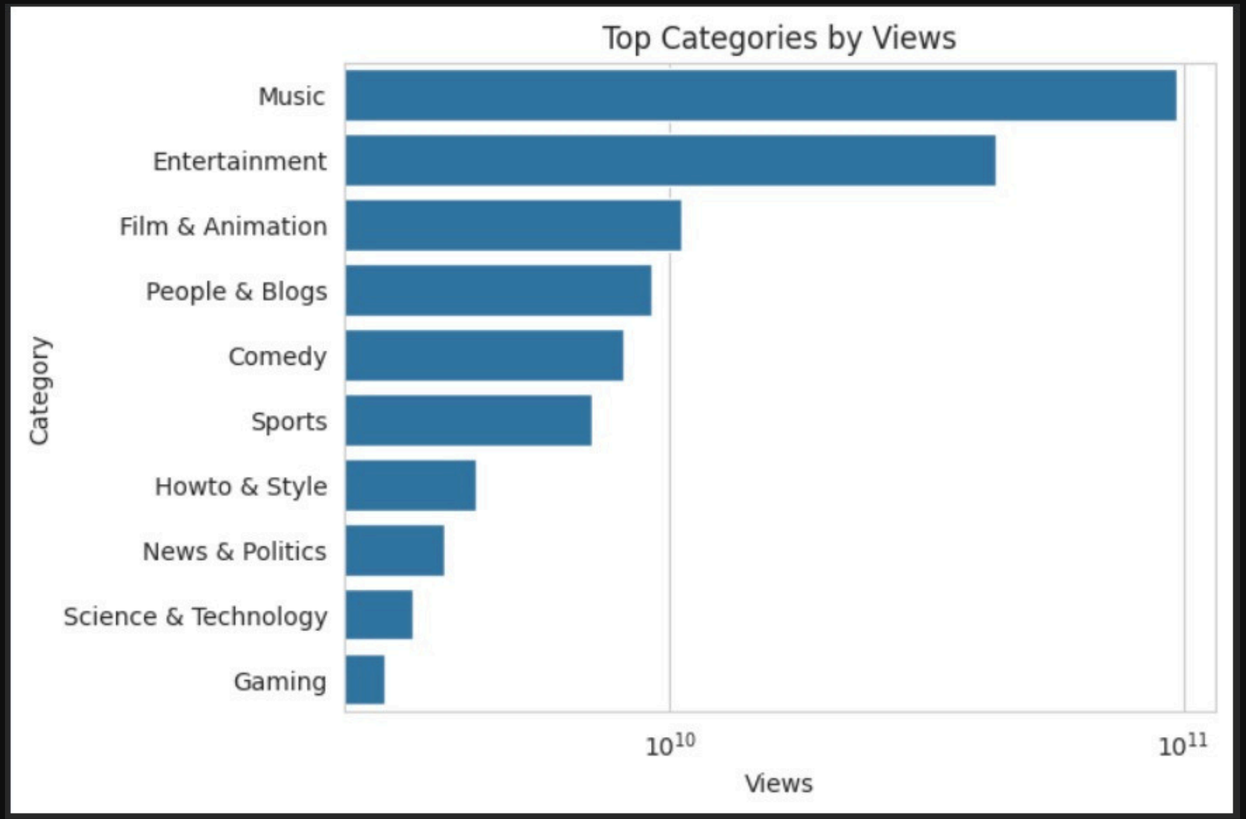
- Entertainment holds the largest share (~40%+)
- People & Blogs and Music follow as major contributors
- Other categories have significantly smaller market share

```
df['category_name'] = df['category_id'].map(category_map)
```

```
category_counts = df['category_name'].value_counts().head(5)

plt.figure()
plt.pie(category_counts, labels=category_counts.index, autopct='%1.1f%%')
plt.title("Top 5 Categories Distribution")
plt.show()
```

Top Categories by Views

- Music dominates total views by a large margin
- Entertainment is the second highest performing category
- Gaming and niche categories have lower overall reach




```python
# map category names
import json

with open("/content/youtube_dataset/IN_category_id.json") as f:
    data = json.load(f)

category_map = {int(i['id']): i['snippet']['title'] for i in data['items']}

df['category_name'] = df['category_id'].map(category_map)

# plot
top_categories = df.groupby('category_name')['views'].sum().sort_values(ascending=False).head(10)

plt.figure()
sns.barplot(x=top_categories.values, y=top_categories.index)
plt.xscale('log')
plt.title("Top Categories by Views")
plt.xlabel("Views")
plt.ylabel("Category")
plt.show()
```

Key Insights

- Music & Entertainment dominate views; niche categories drive engagement
- Peak time: 9–10 AM & evenings
- Friday performs best; weekends lag
- Likes & comments strongly correlate
- Top creators drive most engagement

Thank you!